



BANGLADESH
UNIVERSITY OF
ENGINEERING &
TECHNOLOGY

DEPARTMENT OF
ELECTRICAL &
ELECTRONIC
ENGINEERING

EEE 263
Electronic Circuits

Oscillators

Khairul Islam

Lecturer

Department of EEE, BUET

Office: ECE 913A, ECE Building

Email: khairul@eee.buet.ac.bd

Website: khairul.buet.ac.bd

The Oscillator Feedback Loop

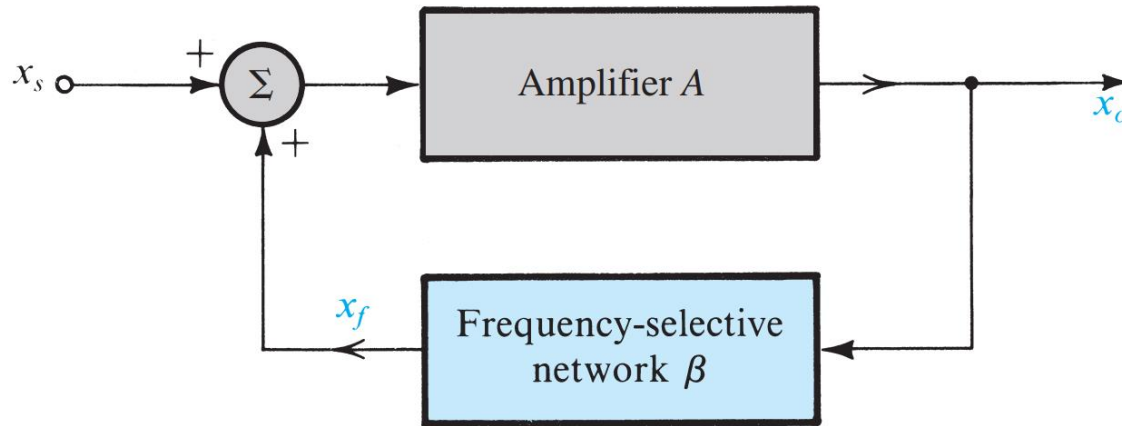


Figure 18.1 The basic structure of a sinusoidal oscillator. A positive-feedback loop is formed by an amplifier and a frequency-selective network. In an actual oscillator circuit, no input signal will be present; here an input signal x_s is employed to help explain the principle of operation.

$$A_f(s) = \frac{A(s)}{1 - A(s)\beta(s)}$$

where we note the negative sign in the denominator. The loop gain $L(s)$ is given by

$$L(s) \equiv A(s)\beta(s)$$

and the characteristic equation is

$$1 - L(s) = 0$$

The Oscillation Criterion



If at a specific frequency f_0 the loop gain $A\beta$ is equal to unity, it follows from Eq. (18.1) that A_f will be infinite. That is, at this frequency the circuit will have a finite output for zero input signal. Such a circuit is by definition an oscillator. Thus the condition for the feedback loop of Fig. 18.1 to provide sinusoidal oscillations of frequency ω_0 is

$$L(j\omega_0) \equiv A(j\omega_0)\beta(j\omega_0) = 1 \quad (18.4)$$

That is, *at ω_0 the phase of the loop gain should be zero and the magnitude of the loop gain should be unity*. This is known as the **Barkhausen criterion**. Note that for the circuit to oscillate at one frequency, the oscillation criterion should be satisfied only at one frequency (i.e., ω_0); otherwise the resulting waveform will not be a simple sinusoid.

The Oscillation Criterion



An intuitive feeling for the Barkhausen criterion can be gained by considering once more the feedback loop of Fig. 18.1. For this loop to *produce* and *sustain* an output x_o with no input applied ($x_s = 0$), the feedback signal x_f ,

$$x_f = \beta x_o$$

should be sufficiently large that when multiplied by A it produces x_o , that is,

$$Ax_f = x_o$$

that is,

$$A\beta x_o = x_o$$

which results in

$$A\beta = 1$$

It should be noted that the *frequency of oscillation* ω_0 is determined solely by the phase characteristics of the feedback loop; the loop oscillates at the frequency for which the phase is zero (or, equivalently, 360°).

Analysis of a given oscillator circuit to determine the frequency of oscillation and the condition for the oscillations to start proceeds in three steps:

1. Break the feedback loop to determine the loop gain $A(s)\beta(s)$. This step is similar to that utilized in Section 11.2 in the analysis of negative-feedback amplifiers.
2. The oscillation frequency ω_0 is found as the frequency for which the phase angle of $A(j\omega)\beta(j\omega)$ is zero or, equivalently, 360° .
3. The condition for the oscillations to start is found from

$$|A(j\omega_0)\beta(j\omega_0)| \geq 1$$

Note that making the magnitude of the loop gain slightly greater than unity ensures that oscillations will start.

Example

Figure 18.3(a) shows a sinusoidal oscillator formed by placing a second-order LCR bandpass filter [see Fig. 17.18(d)] in the feedback path of a positive-gain amplifier. Find the frequency of oscillation ω_0 , and the condition for oscillations to start. Assume an ideal op amp.

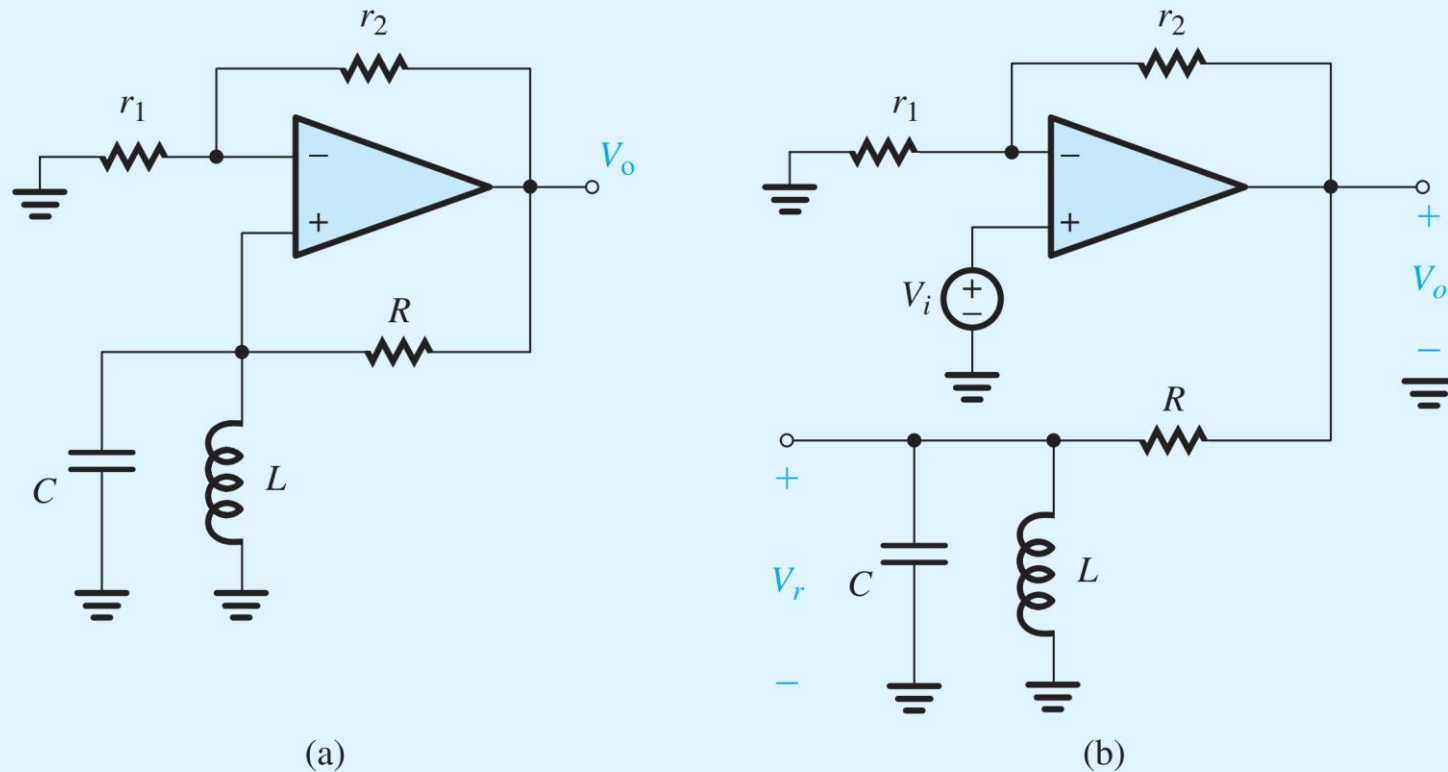


Figure 18.3 (a) An oscillator formed by connecting a positive-gain amplifier in a feedback loop with a bandpass RLC circuit. (b) Breaking the feedback loop at the input of the op amp to determine $A(s) \equiv V_o(s)/V_i(s)$ and $\beta(s) \equiv V_r(s)/V_o(s)$, and hence the loop gain $A(s)\beta(s)$.

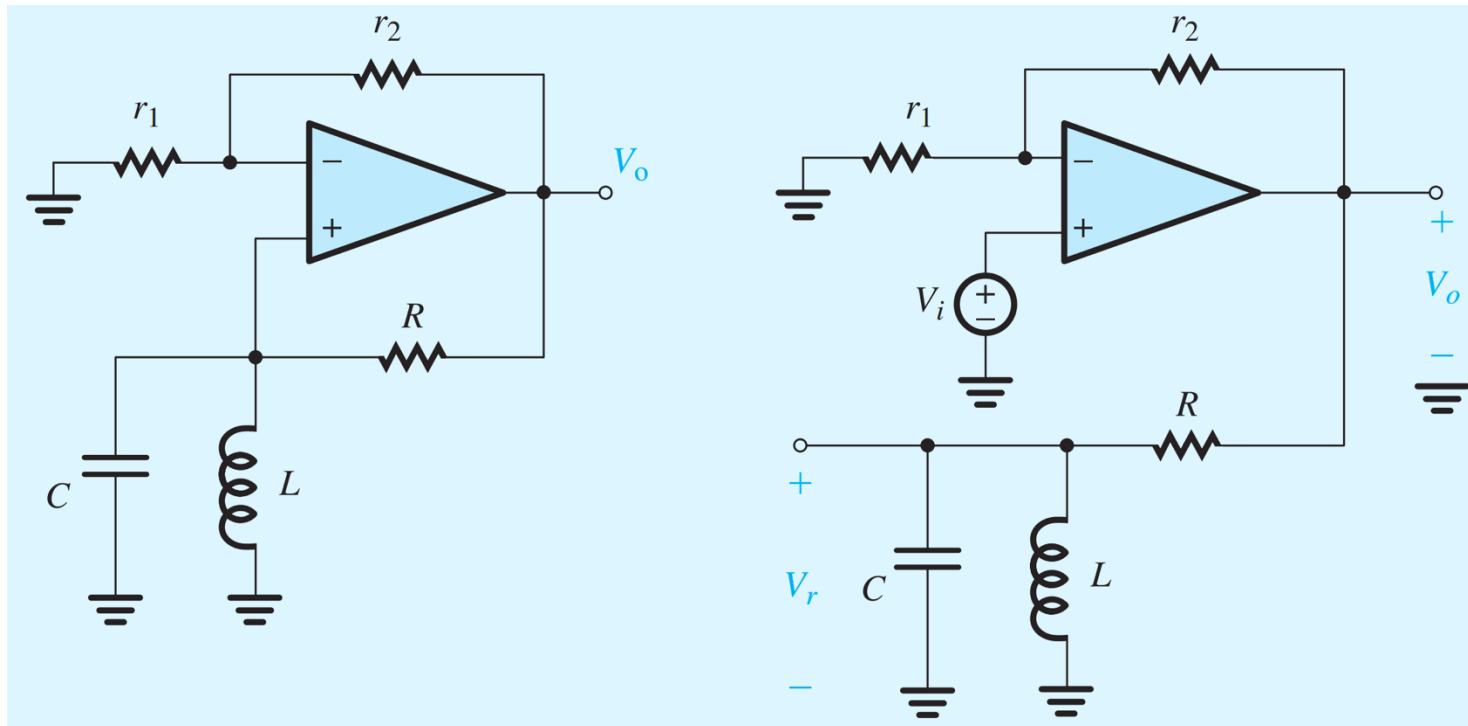
$$A(s) = 1 + \frac{r_2}{r_1}$$

$$\frac{1}{sC} || sL = \frac{\frac{1}{sC} \times sL}{\frac{1}{sC} + sL} = \frac{\frac{L}{C}}{\frac{1 + s^2 LC}{sC}}$$

$$\Rightarrow \frac{1}{sC} || sL = \frac{sL}{1 + s^2 LC}$$

$$\therefore \beta(s) = \frac{V_r(s)}{V_o(s)} = \frac{\frac{1}{sC} || sL}{R + \frac{1}{sC} || sL}$$

Example



$$\therefore A(s)\beta(s) = \frac{\frac{s}{RC} \times \left(1 + \frac{r_2}{r_1}\right)}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

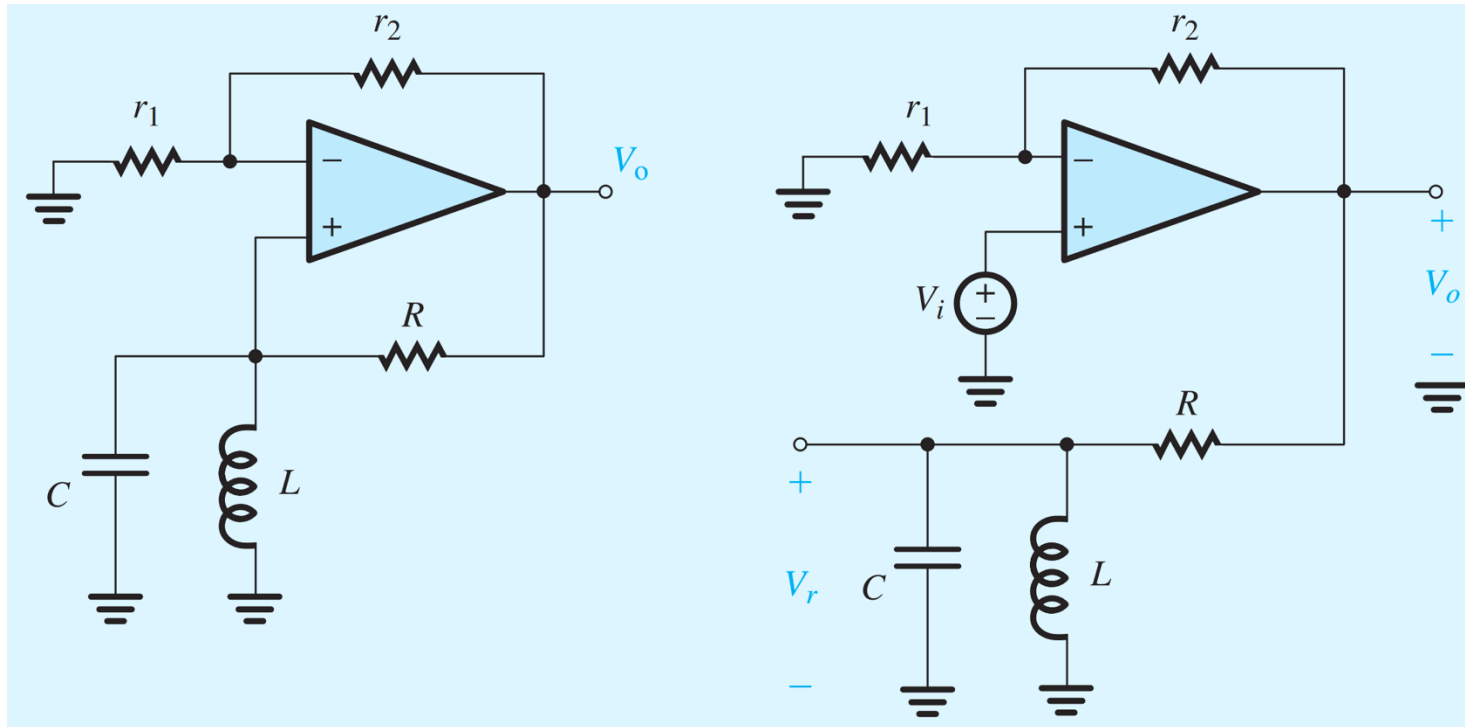
$$\therefore A(j\omega)\beta(j\omega) = \frac{\frac{j\omega}{RC} \times \left(1 + \frac{r_2}{r_1}\right)}{(j\omega)^2 + \frac{j\omega}{RC} + \frac{1}{LC}}$$

$$\therefore \beta(s) = \frac{V_r(s)}{V_o(s)} = \frac{\frac{1}{sC} \parallel sL}{R + \frac{1}{sC} \parallel sL} = \frac{\frac{sL}{1 + s^2LC}}{R + \frac{sL}{1 + s^2LC}}$$

$$\Rightarrow \beta(s) = \frac{sL}{s^2RLC + sL + R} = \frac{\frac{s}{RC}}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$\therefore A(j\omega)\beta(j\omega) = \frac{\frac{j\omega}{RC} \times \left(1 + \frac{r_2}{r_1}\right)}{\left(-\omega^2 + \frac{1}{LC}\right) + \frac{j\omega}{RC}}$$

Example



$$\therefore A(j\omega)\beta(j\omega) = \frac{\frac{j\omega}{RC} \times \left(1 + \frac{r_2}{r_1}\right)}{\left(-\omega + \frac{1}{LC}\right)^2 + \frac{j\omega}{RC}}$$

The phase angle of $A(j\omega)\beta(j\omega)$ will be zero at the value of ω that makes the real part of the denominator zero, thus

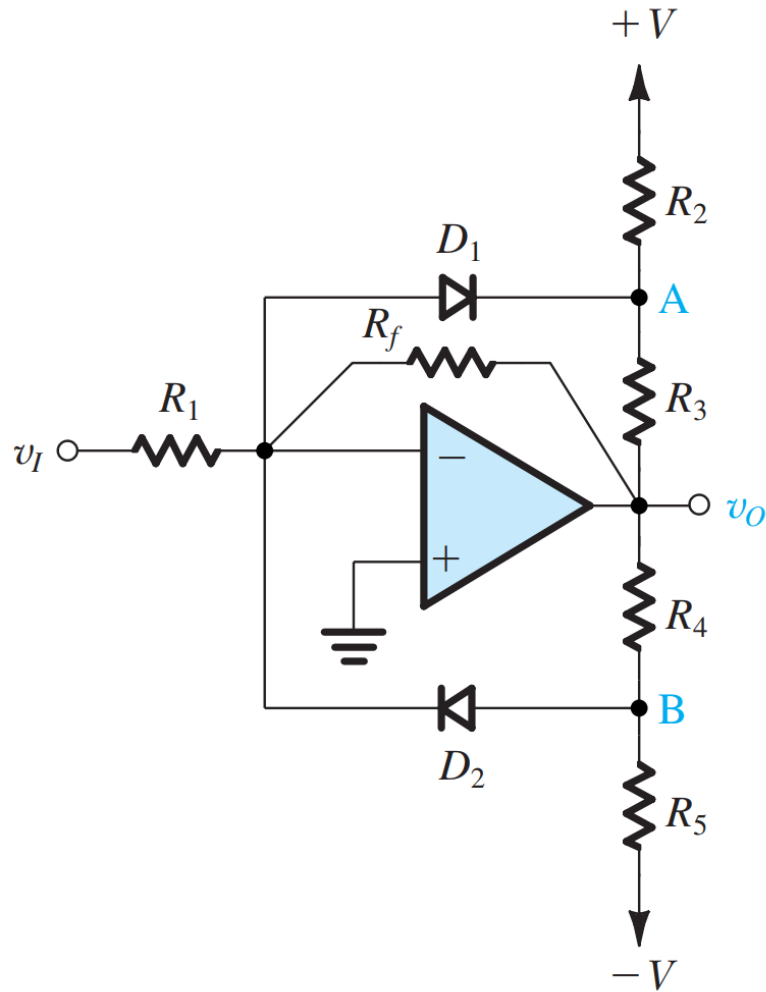
$$\omega_0 = \sqrt{\frac{1}{LC}}$$

At this frequency, the magnitude of the loop gain is given by

$$|A(j\omega_0)\beta(j\omega_0)| = \left(1 + \frac{r_2}{r_1}\right) \geq 1$$

$$\therefore \frac{r_2}{r_1} \geq 0$$

Amplitude Control



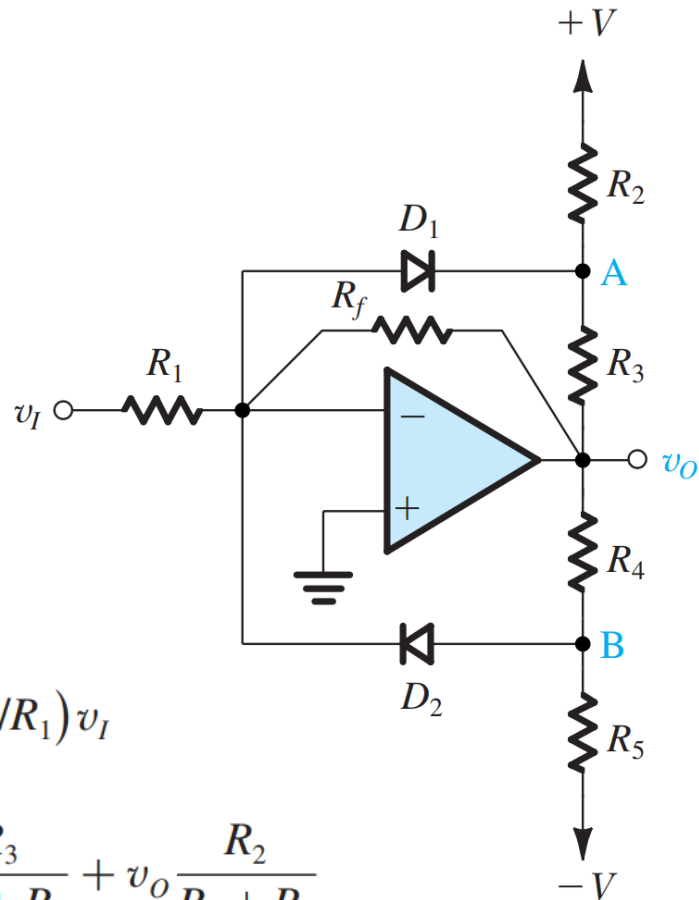
$$v_O = -(R_f/R_1)v_I$$

When D_1 and D_2 are OFF

$$v_A = V \frac{R_3}{R_2 + R_3} + v_O \frac{R_2}{R_2 + R_3}$$

$$v_B = -V \frac{R_4}{R_4 + R_5} + v_O \frac{R_5}{R_4 + R_5}$$

Amplitude Control



$$v_O = -(R_f/R_1)v_I$$

$$v_A = V \frac{R_3}{R_2 + R_3} + v_O \frac{R_2}{R_2 + R_3}$$

$$v_B = -V \frac{R_4}{R_4 + R_5} + v_O \frac{R_5}{R_4 + R_5}$$

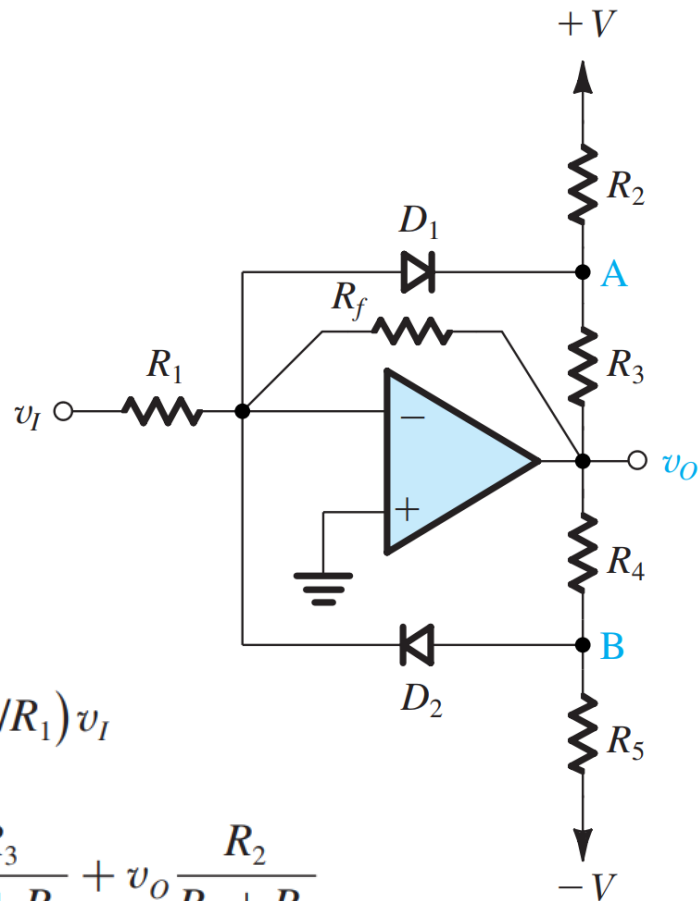
As v_I goes positive, v_O goes negative. v_B will become more negative, thus keeping D_2 off. However, v_A becomes less positive. Then, if we continue to increase v_I , a negative value of v_O will be reached at which v_A becomes -0.7 V or so and diode D_1 conducts.

The value of v_O at which D_1 conducts is the **negative limiting level**, which we denote L_- ,

$$L_- = -V \frac{R_3}{R_2} - V_D \left(1 + \frac{R_3}{R_2} \right)$$

The corresponding value of v_I can be found by dividing L_- by the limiter gain $-R_f/R_1$.

Amplitude Control



$$v_O = -(R_f/R_1)v_I$$

$$v_A = V \frac{R_3}{R_2 + R_3} + v_O \frac{R_2}{R_2 + R_3}$$

$$v_B = -V \frac{R_4}{R_4 + R_5} + v_O \frac{R_5}{R_4 + R_5}$$

As v_I goes positive, v_O goes negative. v_B will become more negative, thus keeping D_2 off. However, v_A becomes less positive. Then, if we continue to increase v_I , a negative value of v_O will be reached at which v_A becomes -0.7 V or so and diode D_1 conducts.

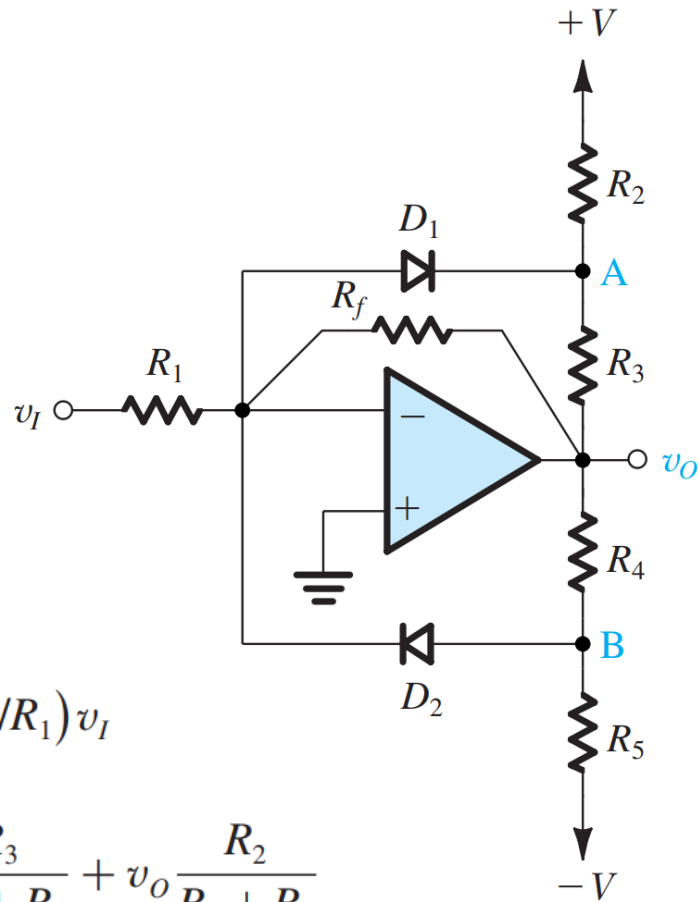
The value of v_O at which D_1 conducts is the **negative limiting level**, which we denote L_- ,

$$L_- = -V \frac{R_3}{R_2} - V_D \left(1 + \frac{R_3}{R_2} \right)$$

The corresponding value of v_I can be found by dividing L_- by the limiter gain $-R_f/R_1$.

If v_I is increased beyond this value, more current is injected into D_1 , and v_A remains at approximately $-V_D$. Thus, the current through R_2 remains constant, and the additional diode current flows through R_3 . Thus, R_3 appears in effect in parallel with R_f , and the incremental gain (ignoring the diode resistance) is $-(R_f \times R_3)/R_1$.

Amplitude Control



$$v_O = -(R_f/R_1)v_I$$

$$v_A = V \frac{R_3}{R_2 + R_3} + v_O \frac{R_2}{R_2 + R_3}$$

$$v_B = -V \frac{R_4}{R_4 + R_5} + v_O \frac{R_5}{R_4 + R_5}$$

The transfer characteristic for negative v_I can be found in a manner identical to that just employed. It can be easily seen that for negative v_I , diode D_2 plays an identical role to that played by diode D_1 for positive v_I .

The value of v_O at which D_2 conducts is the **Positive limiting level**, which we denote L_+ ,

$$L_+ = V \frac{R_4}{R_5} + V_D \left(1 + \frac{R_4}{R_5} \right)$$

The slope of the transfer characteristic in the positive limiting region is $-(R_f \times R_4)/R_1$.

Amplitude Control

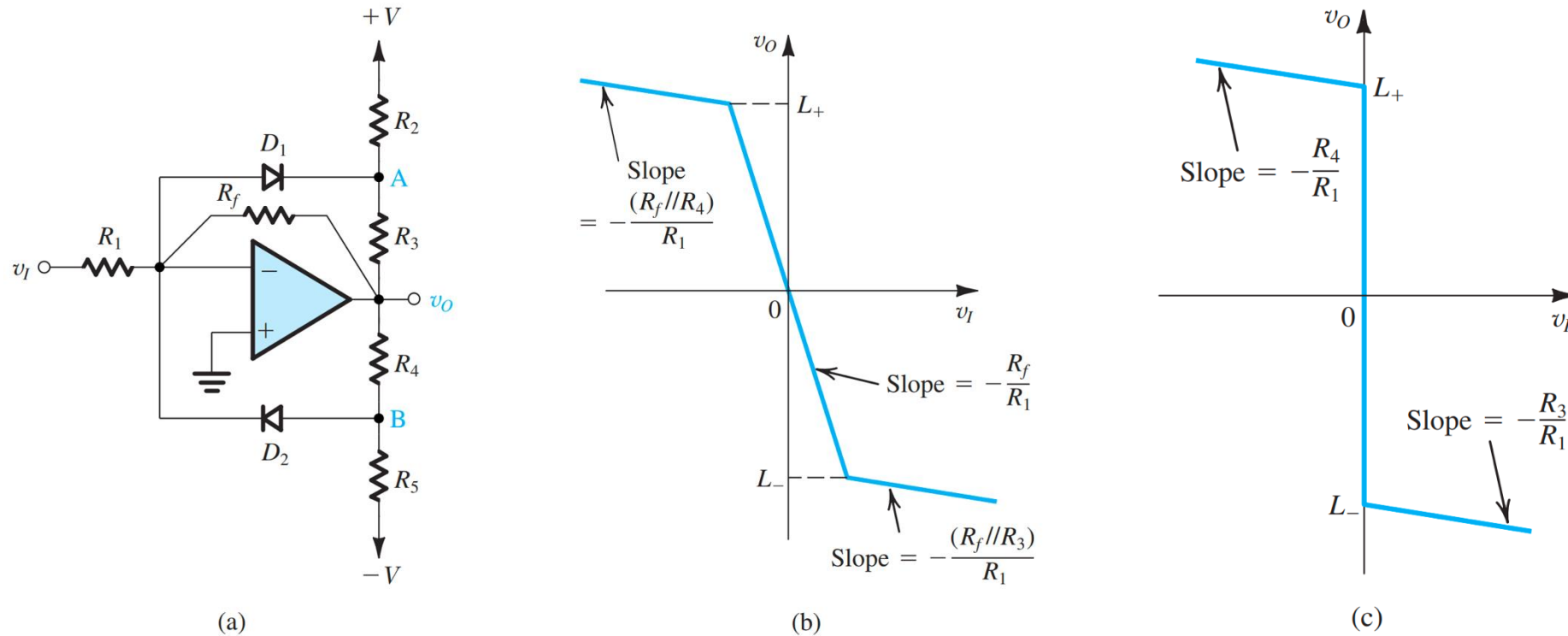
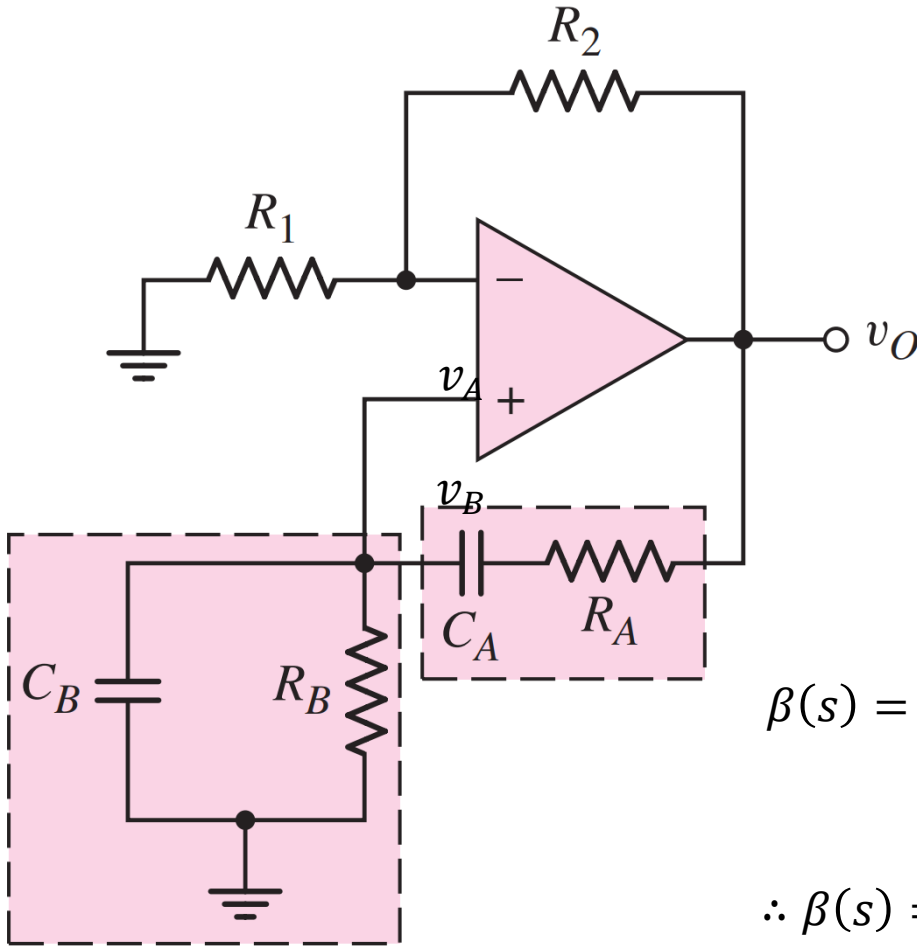


Figure 18.4 (a) A popular limiter circuit. (b) Transfer characteristic of the limiter circuit; L_- and L_+ are given by Eqs. (18.8) and (18.9), respectively. (c) When R_f is removed, the limiter turns into a comparator with the characteristic shown.

Wien Bridge Oscillator



$$A(s) = 1 + \frac{R_2}{R_1}$$

$$Z_p = R_B \parallel \frac{1}{sC_B} = \frac{R_B \times \frac{1}{sC_B}}{R_B + \frac{1}{sC_B}} = \frac{R_B}{1 + sR_B C_B}$$

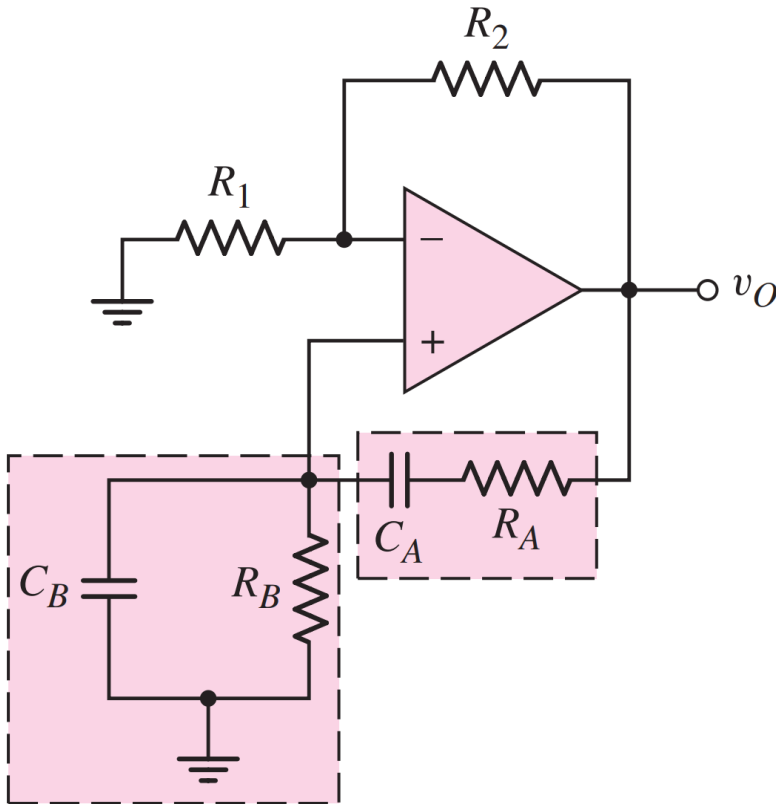
$$Z_s = R_A + \frac{1}{sC_A} = \frac{1 + sR_A C_A}{sC_A}$$

$$\beta(s) = \frac{Z_p}{Z_p + Z_s} = \frac{\frac{R_B}{1 + sR_B C_B}}{\frac{R_B}{1 + sR_B C_B} + \frac{1 + sR_A C_A}{sC_A}} = \frac{sR_B C_A}{sR_B C_A + (1 + sR_A C_A)(1 + sR_B C_B)}$$

$$\therefore \beta(s) = \frac{sR_B C_A}{(1 + s^2 R_A R_B C_A C_B) + s(R_A C_A + R_B C_B + R_B C_A)}$$

$$\therefore A(s)\beta(s) = \left(1 + \frac{R_2}{R_1}\right) \times \frac{sR_B C_A}{(1 + s^2 R_A R_B C_A C_B) + s(R_A C_A + R_B C_B + R_B C_A)}$$

Wien Bridge Oscillator



$$\therefore A(j\omega)\beta(j\omega) = \left(1 + \frac{R_2}{R_1}\right) \times \frac{j\omega R_B C_A}{(1 - \omega^2 R_A R_B C_A C_B) + j\omega(R_A C_A + R_B C_B + R_B C_A)}$$

The phase angle of $A(j\omega)\beta(j\omega)$ will be zero at the value of ω that makes the real part of the denominator zero, thus

$$1 - \omega^2 R_A R_B C_A C_B = 0$$

$$\Rightarrow \omega_0 = \sqrt{\frac{1}{R_A R_B C_A C_B}}$$

At this frequency, the magnitude of the loop gain is given by

$$|A(j\omega_0)\beta(j\omega_0)| = \left(1 + \frac{R_2}{R_1}\right) \times \frac{R_B C_A}{(R_A C_A + R_B C_B + R_B C_A)} \geq 1$$

$$\Rightarrow 1 + \frac{R_2}{R_1} \geq \frac{R_A C_A + R_B C_B + R_B C_A}{R_B C_A} = 1 + \frac{R_A}{R_B} + \frac{C_B}{C_A}$$

$$\therefore \frac{R_2}{R_1} \geq \frac{R_A}{R_B} + \frac{C_B}{C_A}$$

Wien Bridge Oscillator

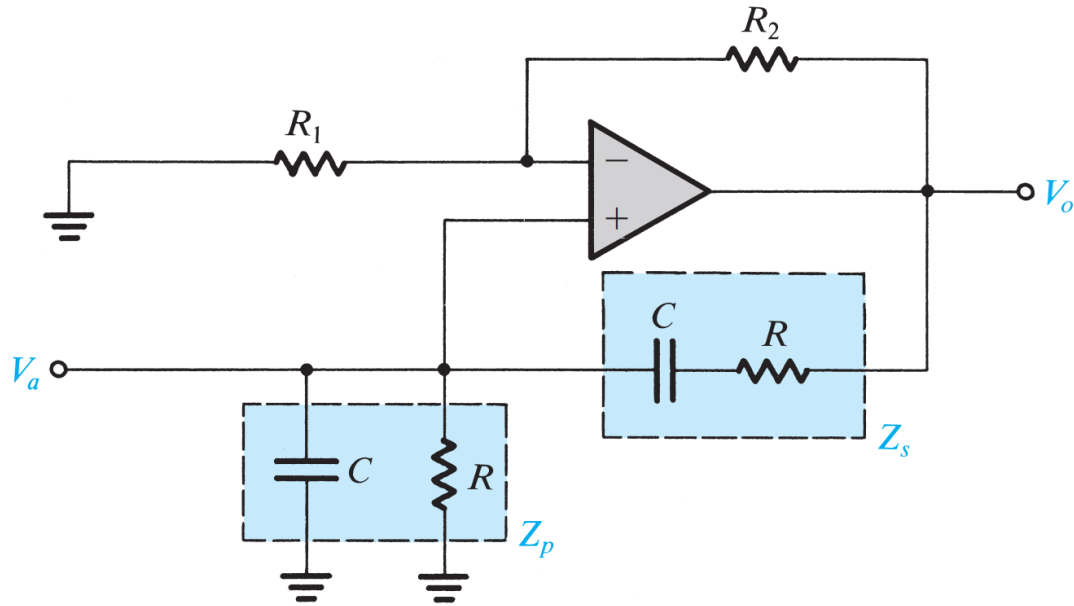


Figure 18.5 A Wien-bridge oscillator without amplitude stabilization.

$$L(s) = \left[1 + \frac{R_2}{R_1} \right] \frac{Z_p}{Z_p + Z_s}$$
$$= \frac{1 + R_2/R_1}{1 + Z_s Y_p}$$

Wien Bridge Oscillator

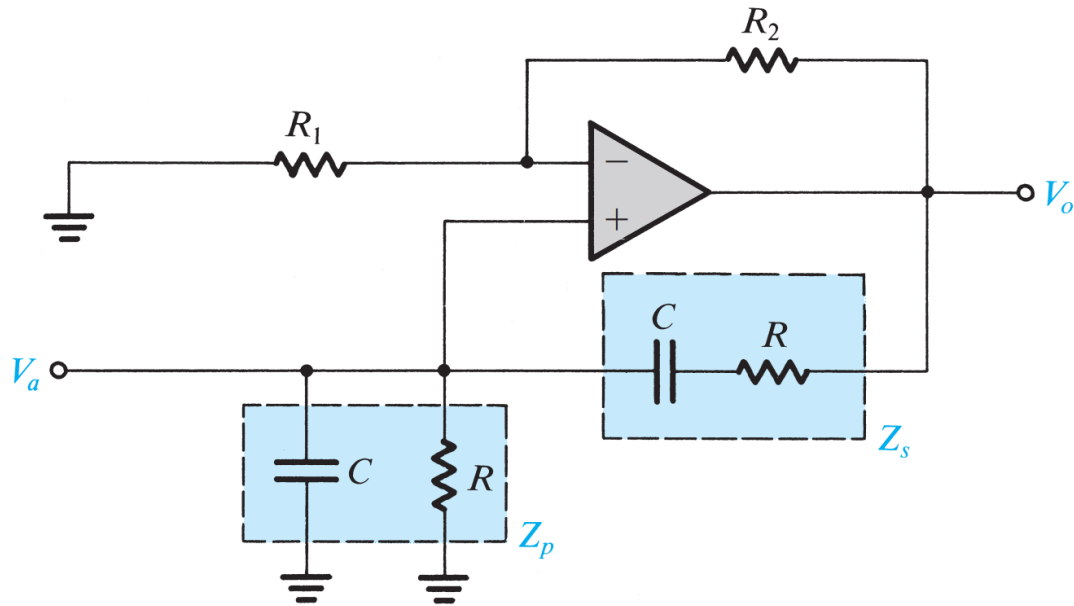


Figure 18.5 A Wien-bridge oscillator without amplitude stabilization.

$$\begin{aligned}
 L(s) &= \left[1 + \frac{R_2}{R_1} \right] \frac{Z_p}{Z_p + Z_s} \\
 &= \frac{1 + R_2/R_1}{1 + Z_s Y_p}
 \end{aligned}$$

$$L(s) = \frac{1 + R_2/R_1}{3 + sCR + 1/sCR}$$

$$L(j\omega) = \frac{1 + R_2/R_1}{3 + j(\omega CR - 1/\omega CR)}$$

$$\omega_0 CR = \frac{1}{\omega_0 CR}$$

$$\omega_0 = 1/CR$$

$$R_2/R_1 = 2$$

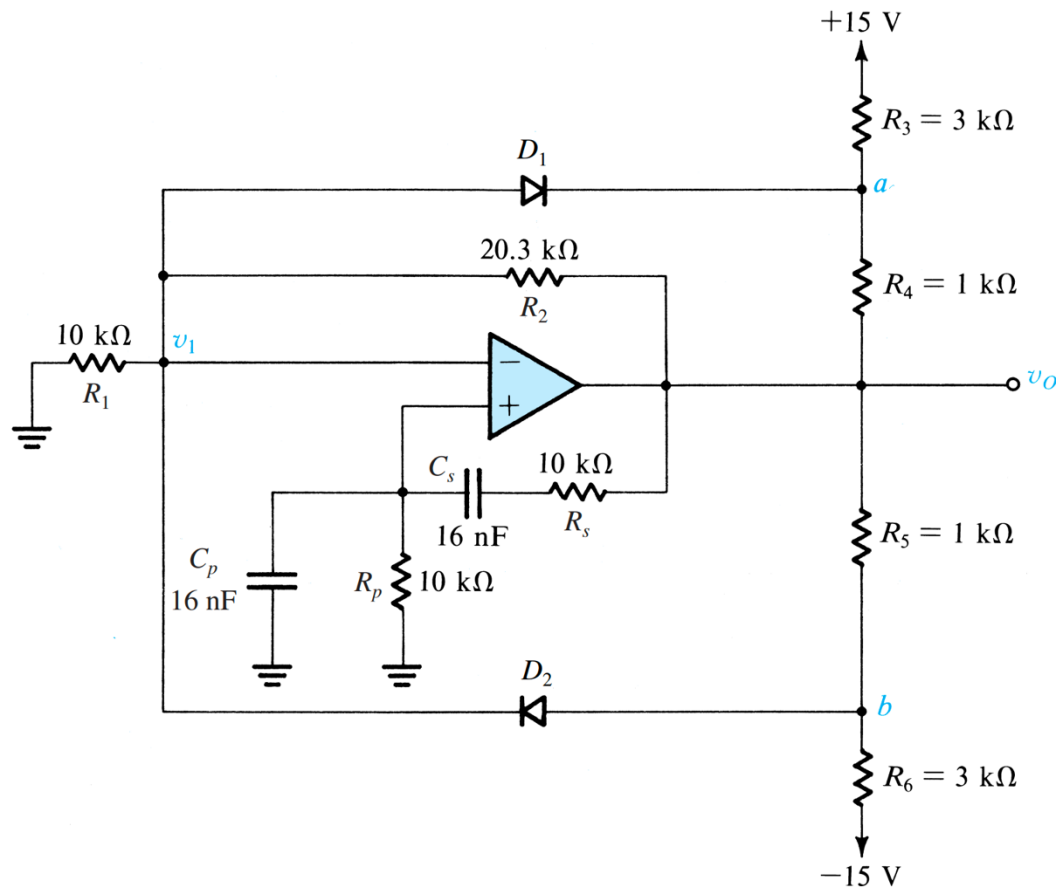
To ensure that oscillations will start, one chooses R_2/R_1 slightly greater than 2.

Wien Bridge Oscillator

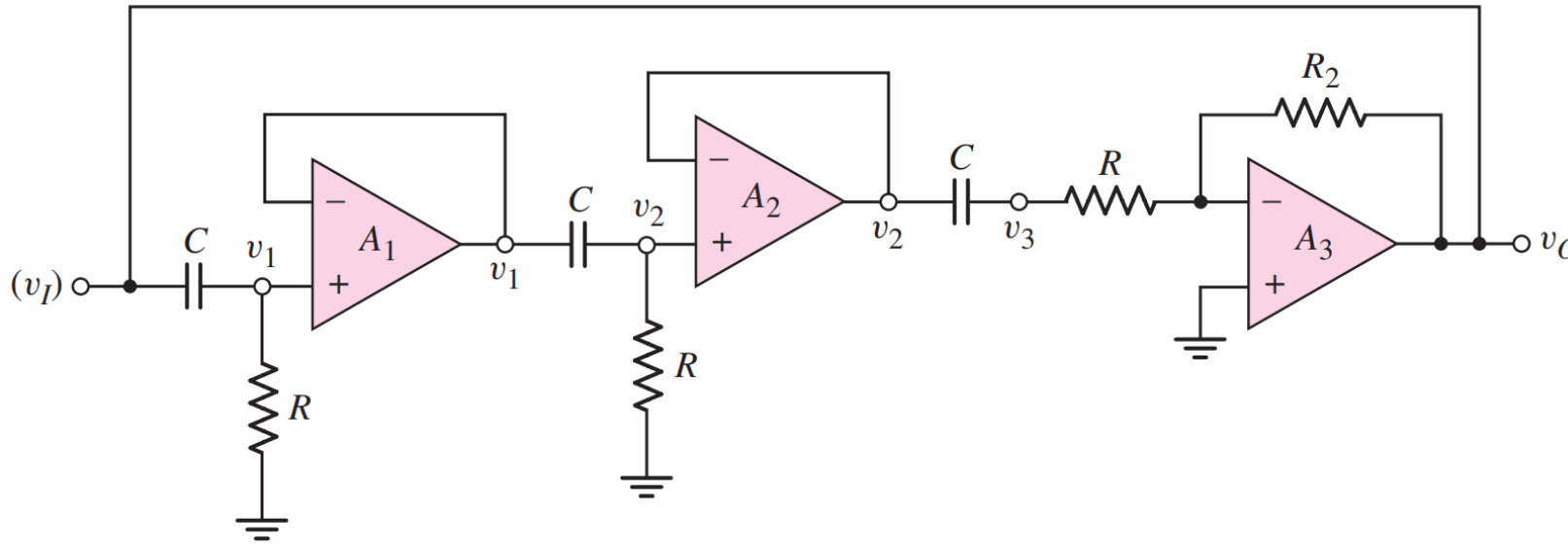


For the circuit in Fig. 18.6: (a) Disregarding the limiter circuit, find the location of the closed-loop poles. (b) Find the frequency of oscillation. (c) With the limiter in place, find the amplitude of the output sine wave (assume that the diode drop is 0.7 V).

Ans. (a) $(10^5/16)(0.015 \pm j)$; (b) 1 kHz; (c) 21.36 V (peak-to-peak)



Phase Shift Oscillator



Phase-shift oscillator circuit with voltage-follower buffer stages

The loop gain is then

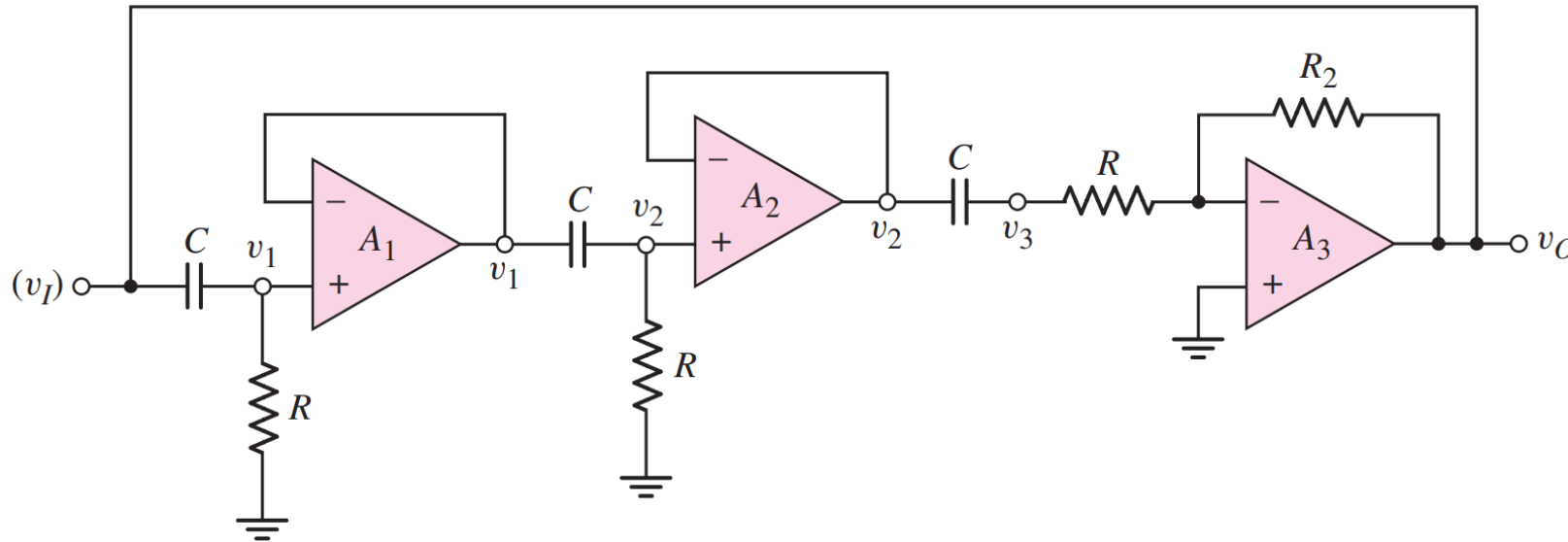
$$T(s) = A(s)\beta(s) = -\frac{R_2}{R} \left(\frac{sRC}{1 + sRC} \right)^3$$

$$\frac{v_1}{v_I} = \frac{R}{R + \frac{1}{sC}} = \frac{sRC}{1 + sRC}$$

$$\therefore \beta(s) = \frac{v_3}{v_I} = \left(\frac{sRC}{1 + sRC} \right)^3$$

$$A(s) = \frac{v_O}{v_3} = -\frac{R_2}{R}$$

Phase Shift Oscillator



Phase-shift oscillator circuit with voltage-follower buffer stages

The condition for oscillation is that

$$|T(j\omega_o)| \geq 1$$

$$\angle T(j\omega_o) = n \times 360^\circ$$

$$T(j\omega) = -\frac{R_2}{R} \left(\frac{j\omega RC}{1 + j\omega RC} \right)^3$$

$$= -\frac{R_2}{R} \times \frac{(j\omega RC)(j\omega RC)^2}{1 + (j\omega RC)^3 + 3 \times 1^2 \times (j\omega RC) + 3 \times 1 \times (j\omega RC)^2}$$

$$\therefore T(j\omega) = -\frac{R_2}{R} \times \frac{-(j\omega RC)(\omega RC)^2}{(1 - 3\omega^2 R^2 C^2) + j\omega RC(3 - \omega^2 R^2 C^2)}$$

Phase Shift Oscillator



$$\therefore T(j\omega) = \frac{R_2}{R} \times \frac{(j\omega RC)(\omega RC)^2}{(1 - 3\omega^2 R^2 C^2) + j\omega RC(3 - \omega^2 R^2 C^2)}$$

For oscillation,

$$\angle T(j\omega_o) = n \times 360^\circ$$

$$1 - 3\omega_o^2 R^2 C^2 = 0$$

$$\therefore \omega_o = \frac{1}{\sqrt{3}RC}$$

At this frequency, the magnitude of the loop gain is given by

$$|T(j\omega_o)| = \frac{R_2}{R} \times \frac{\frac{1}{3}}{3 - \frac{1}{3}} = \frac{R_2}{R} \times \frac{1}{8} \geq 1$$

$$\therefore \frac{R_2}{R} \geq 8$$

Phase Shift Oscillator



$$\frac{j\omega_o RC}{1 + j\omega_o RC} = \frac{(j/\sqrt{3})}{1 + (j/\sqrt{3})} = \frac{j}{\sqrt{3} + j}$$

which can be written in terms of the magnitude and phase, as follows:

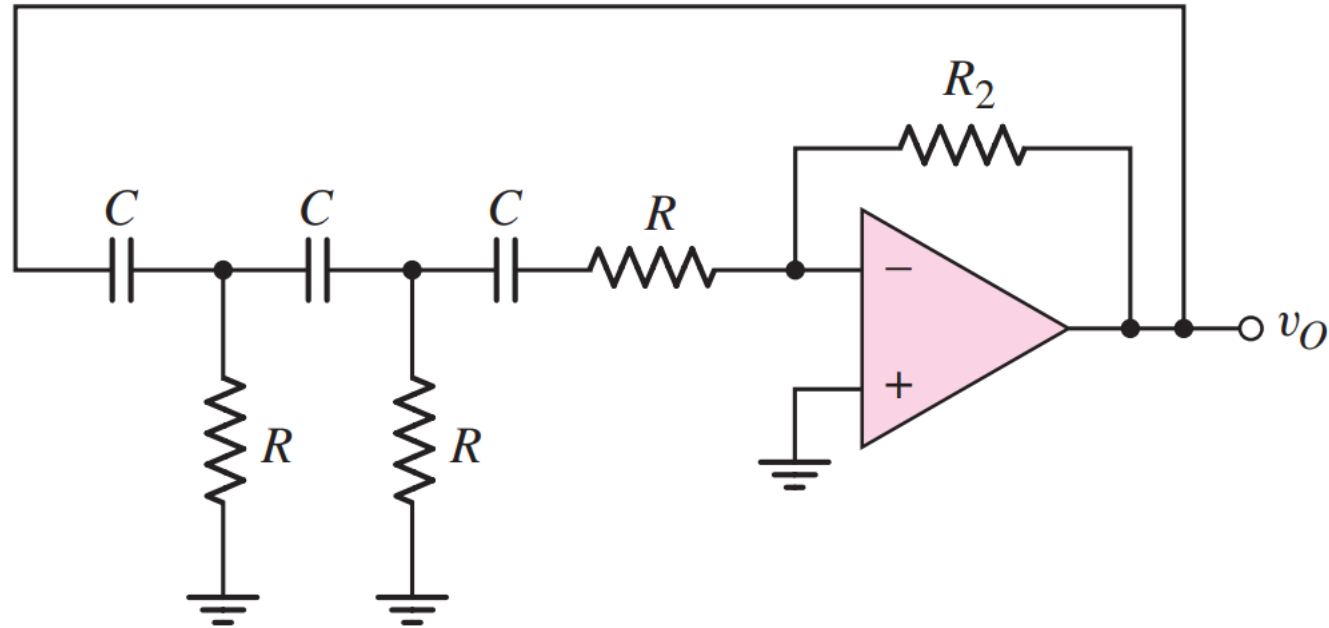
$$\frac{1}{\sqrt{3} + j} \times \frac{\angle 90^\circ}{\angle \tan^{-1}(1/\sqrt{3})} = \frac{1}{2} \times [\angle 90^\circ - \angle \tan^{-1}(0.577)]$$

or

$$\frac{1}{2} \times (\angle 90^\circ - \angle 30^\circ) = \frac{1}{2} \times \angle 60^\circ$$

As required, each RC network introduces a 60 degree phase shift, but they each also introduce an attenuation factor of $(\frac{1}{2})$ for which the amplifier must compensate.

Phase Shift Oscillator

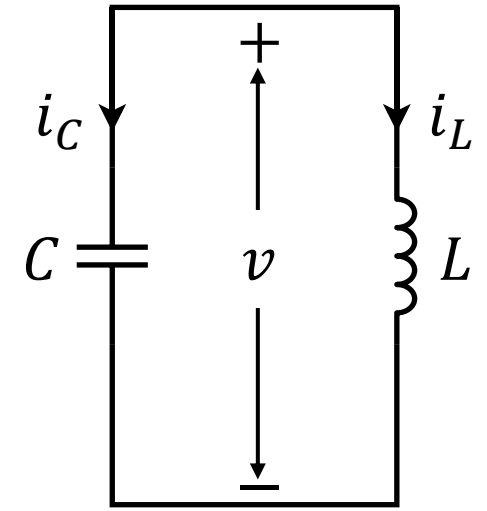
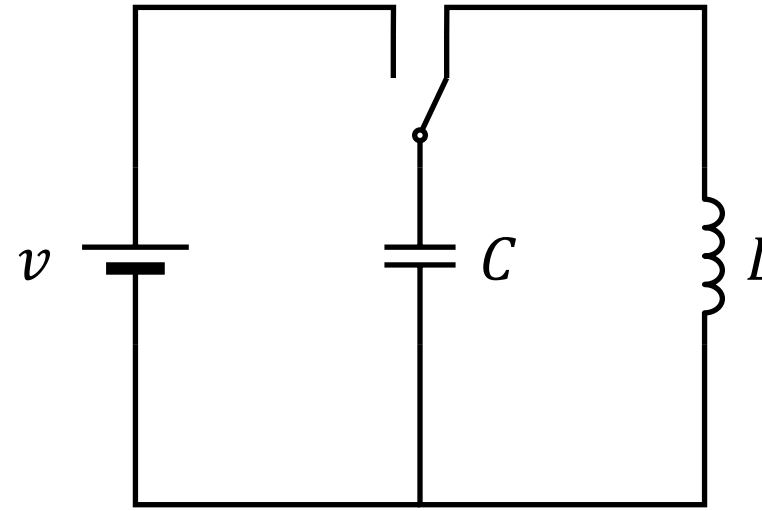
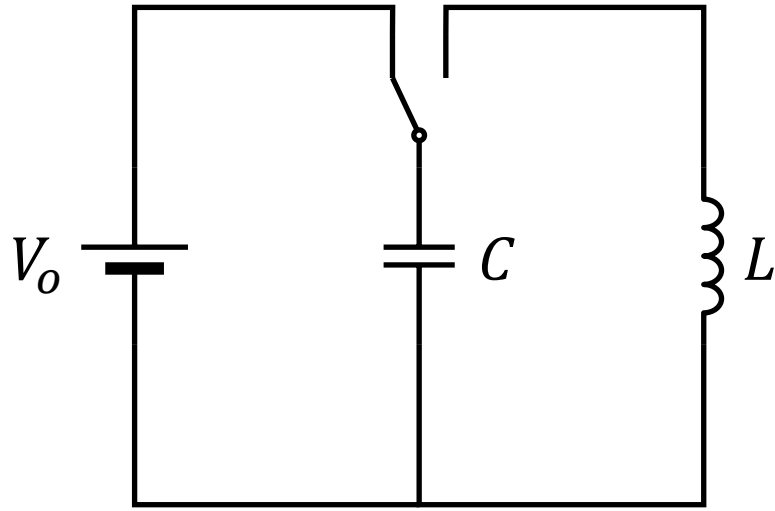


Phase-shift oscillator circuit

$$\omega_o = \frac{1}{\sqrt{6}RC}$$

$$\frac{R_2}{R} = 29$$

LC Resonator



$$\begin{aligned} i_C + i_L &= 0 \\ \Rightarrow \frac{di_C}{dt} + \frac{di_L}{dt} &= 0 \\ \Rightarrow \frac{d}{dt} \left(C \frac{dv}{dt} \right) + \frac{di_L}{dt} &= 0 \end{aligned}$$

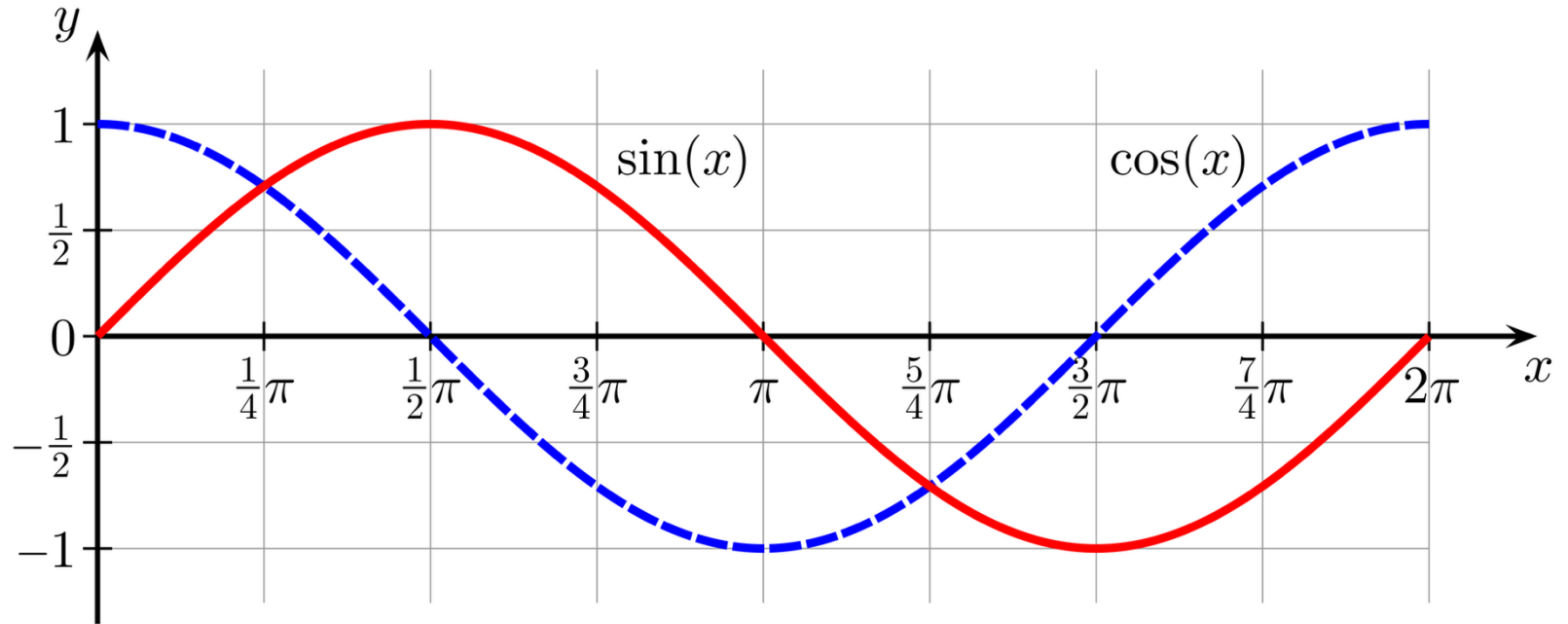
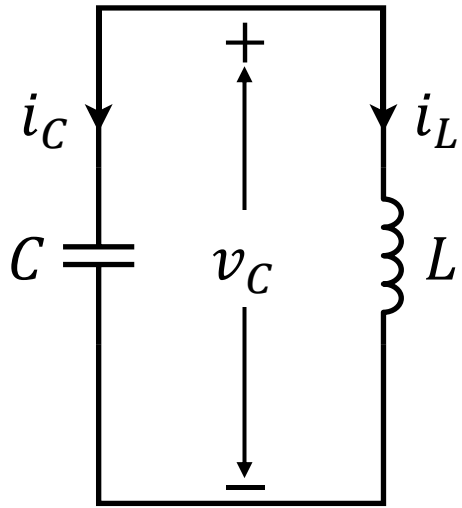
$$\Rightarrow C \frac{d^2 v}{dt^2} + \frac{di_L}{dt} = 0$$

$$\Rightarrow C \frac{d^2 v}{dt^2} + \frac{1}{L} v = 0$$

$$\Rightarrow \frac{d^2 v}{dt^2} + \frac{1}{LC} v = 0$$

$$\Rightarrow \frac{d^2 v}{dt^2} + \omega_o^2 v = 0$$

LC Resonator



$$\frac{d^2 v}{dt^2} + \omega_o^2 v = 0$$

$$\Rightarrow v_C = A' \cos(\omega_o t)$$

$$\Rightarrow i_L = A'' \sin(\omega_o t)$$

$$\omega_o^2 = \frac{1}{LC}$$

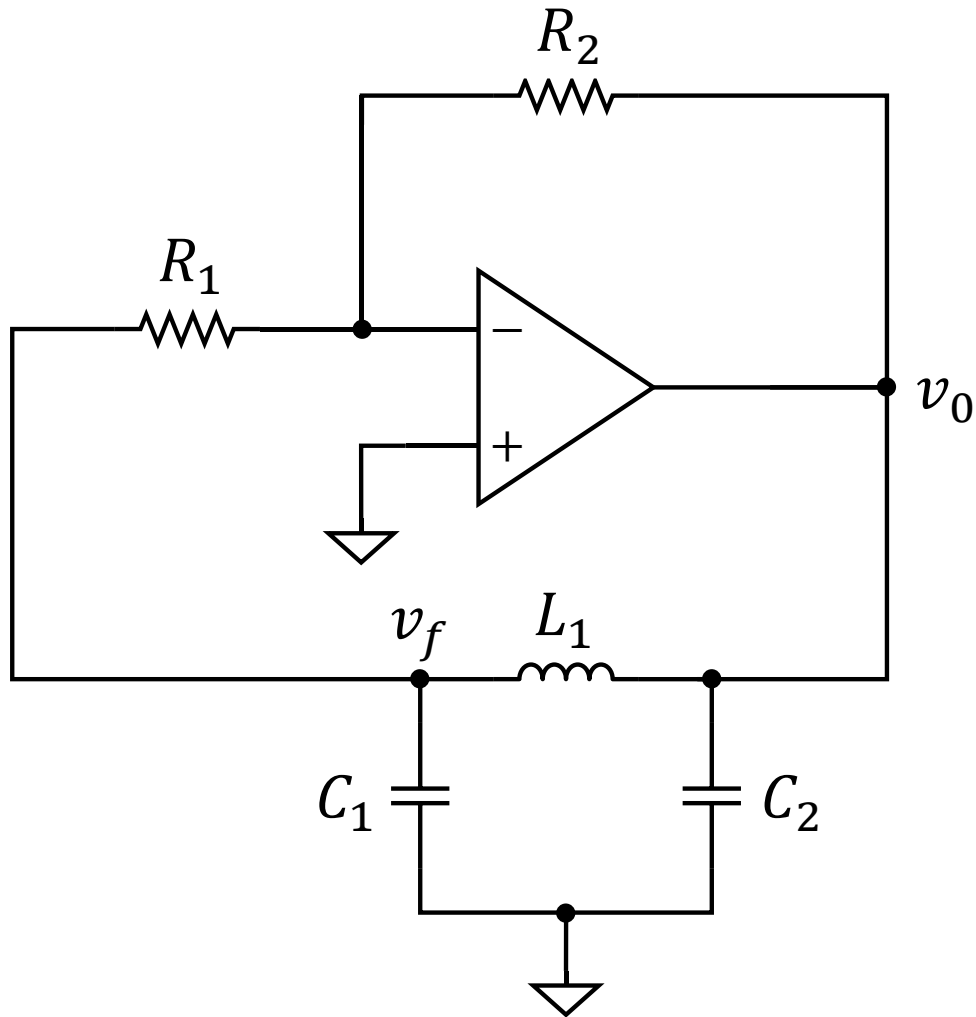
$$\Rightarrow \omega_o = \sqrt{\frac{1}{LC}}$$

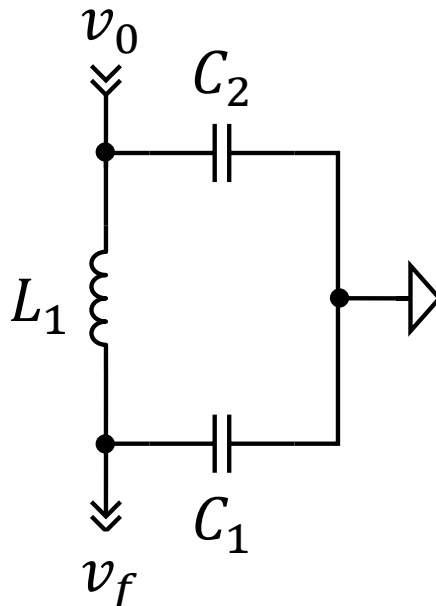
At $t = 0^+$,

$$\frac{1}{2} C V_o^2 = \frac{1}{2} C (A')^2$$

$$\Rightarrow A' = V_o$$

Colpitts Oscillator





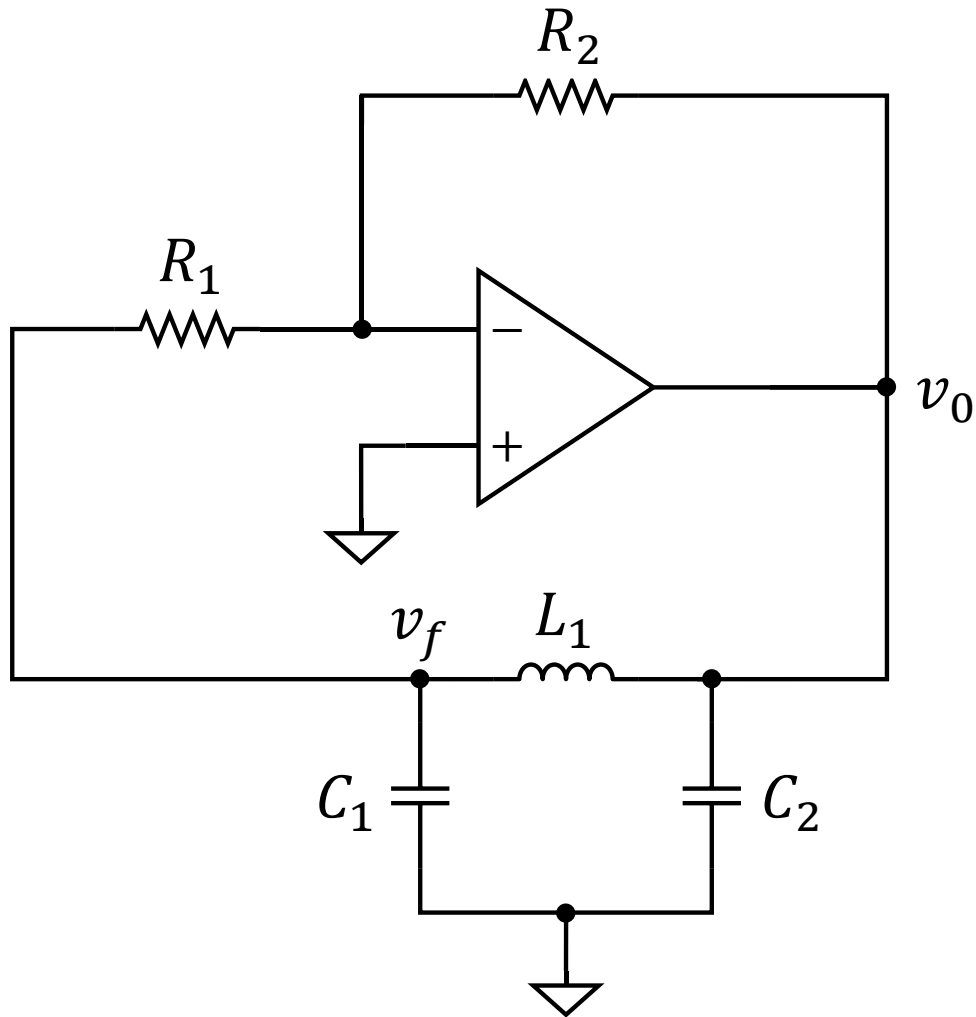
$$\beta = \frac{v_f}{v_0} = -\frac{\frac{1}{sC_1}}{\frac{1}{sC_2}} = -\frac{C_2}{C_1}$$

$$\omega_0 = \sqrt{\frac{1}{L_1 C_T}}$$

Where,

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Colpitts Oscillator



$$\beta = \frac{v_f}{v_o} = -\frac{\frac{1}{sC_1}}{\frac{1}{sC_2}} = -\frac{C_2}{C_1}$$

$$A = -\frac{R_f}{R_1}$$

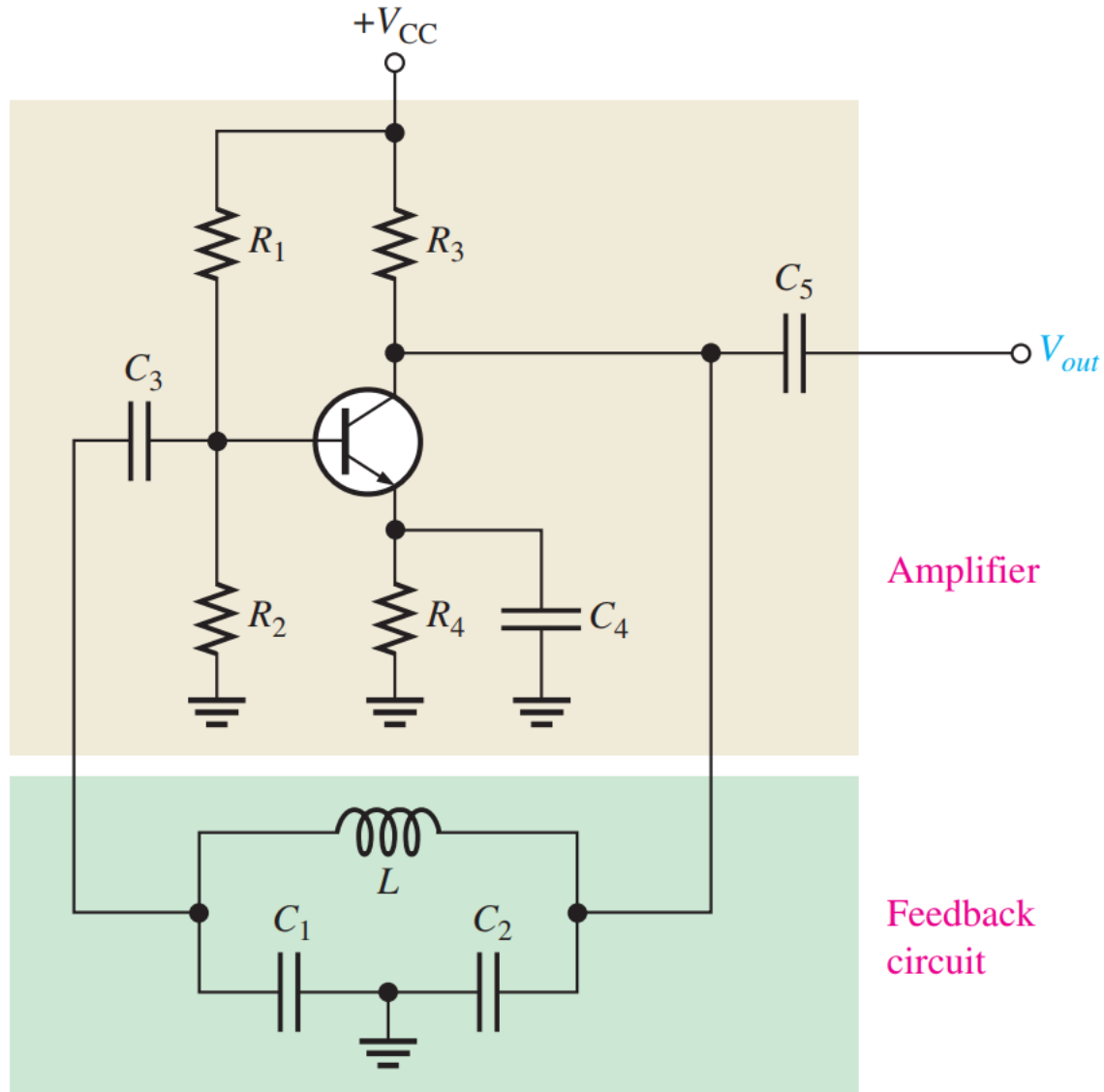
$$A\beta = \frac{R_f}{R_1} \times \frac{C_2}{C_1}$$

For oscillation to start,

$$A\beta = \frac{R_f}{R_1} \times \frac{C_2}{C_1} \geq 1$$

$$\Rightarrow \frac{R_f}{R_1} \geq \frac{C_1}{C_2}$$

Colpitts Oscillator

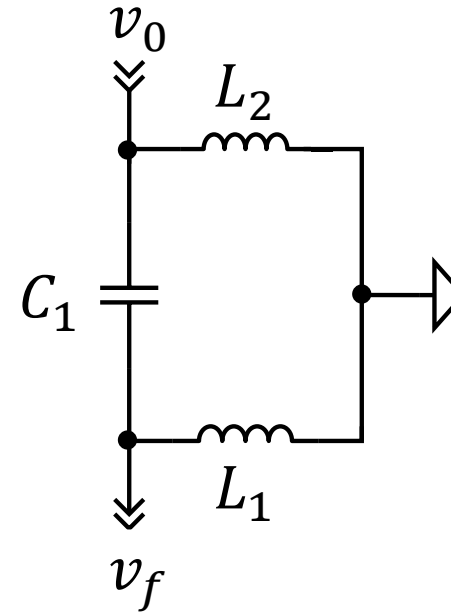
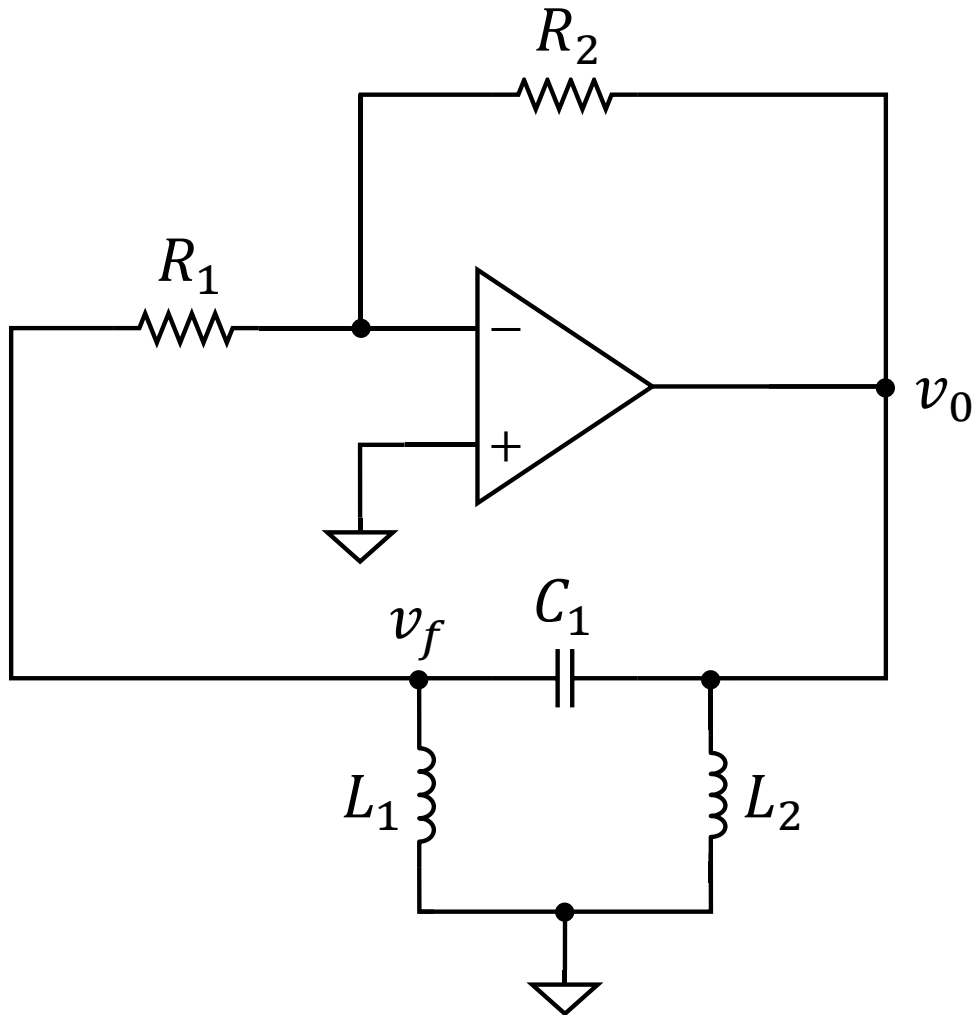


$$f_r \cong \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$A_v > \frac{C_1}{C_2}$$

Hartley Oscillator



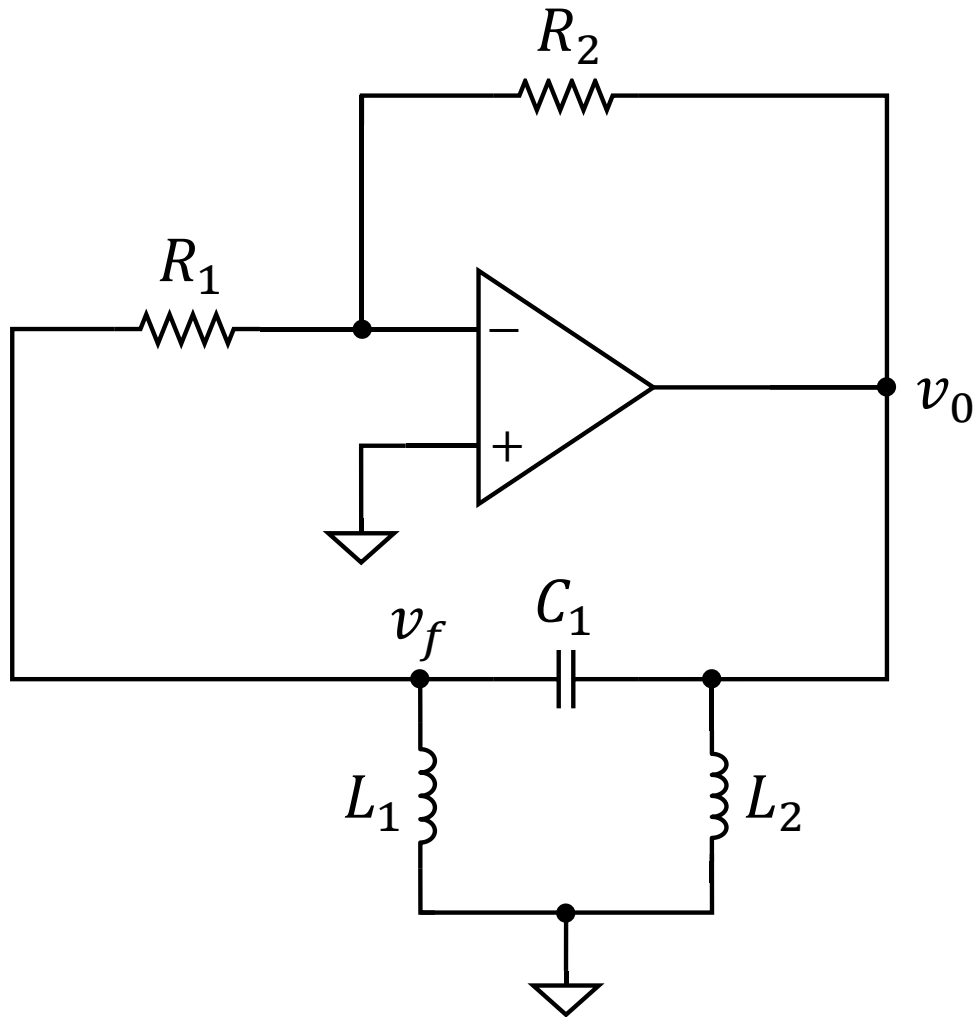
$$\beta = \frac{v_f}{v_0} = -\frac{sL_1}{sL_2} = -\frac{L_1}{L_2}$$

$$\omega_0 = \sqrt{\frac{1}{L_T C_1}}$$

Where,

$$L_T = L_1 + L_2$$

Hartley Oscillator



$$\beta = \frac{v_f}{v_0} = -\frac{sL_1}{sL_2} = -\frac{L_1}{L_2}$$

$$A = -\frac{R_f}{R_1}$$

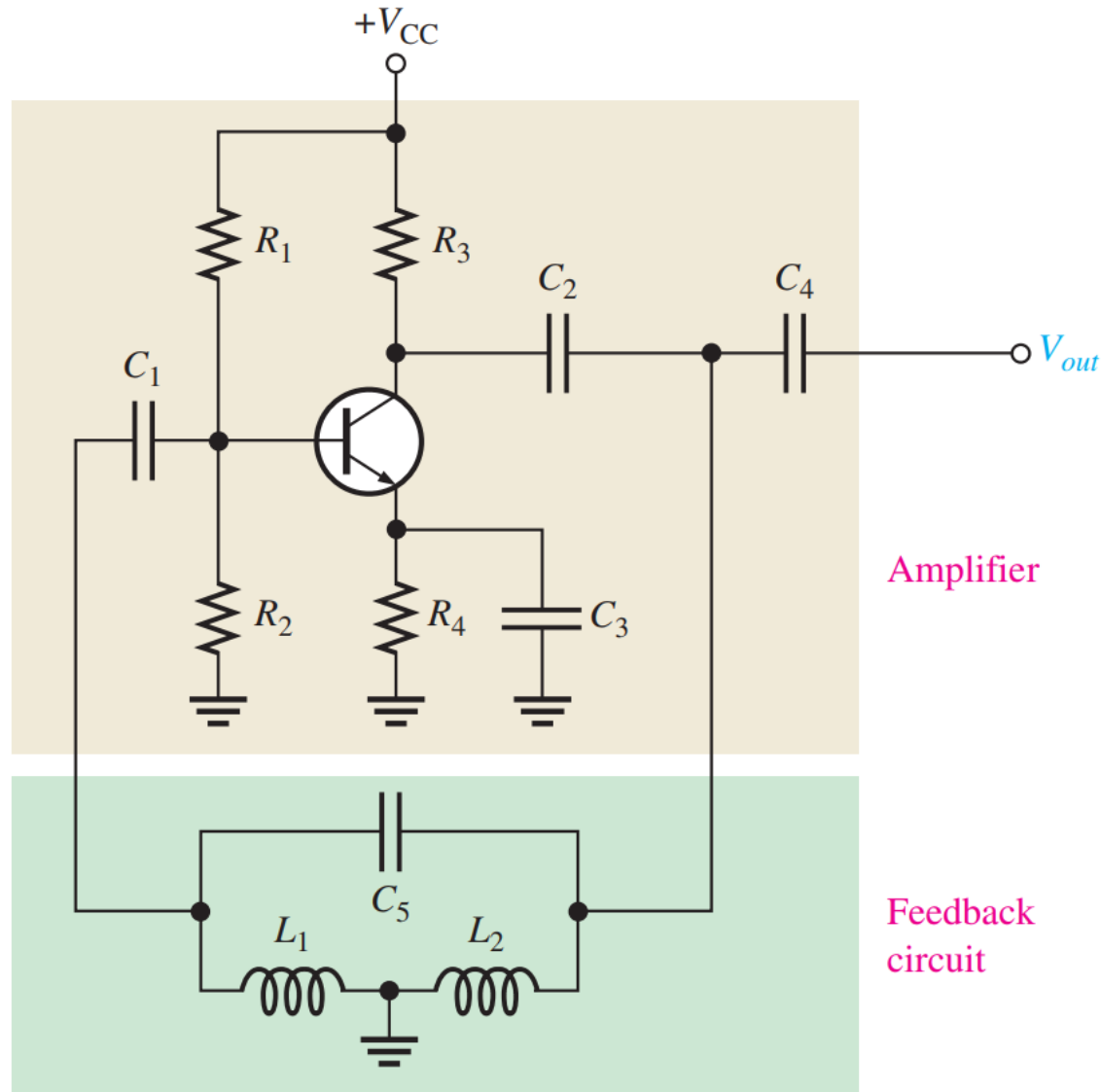
$$A\beta = \frac{R_f}{R_1} \times \frac{L_1}{L_2}$$

For oscillation to start,

$$A\beta = \frac{R_f}{R_1} \times \frac{L_1}{L_2} \geq 1$$

$$\Rightarrow \frac{R_f}{R_1} \geq \frac{L_2}{L_1}$$

Hartley Oscillator



Amplifier

Feedback
circuit

$$f_r \cong \frac{1}{2\pi\sqrt{L_T C}}$$

$$L_T = L_1 + L_2$$

$$A_v \cong \frac{L_2}{L_1}$$

Crystal Oscillator

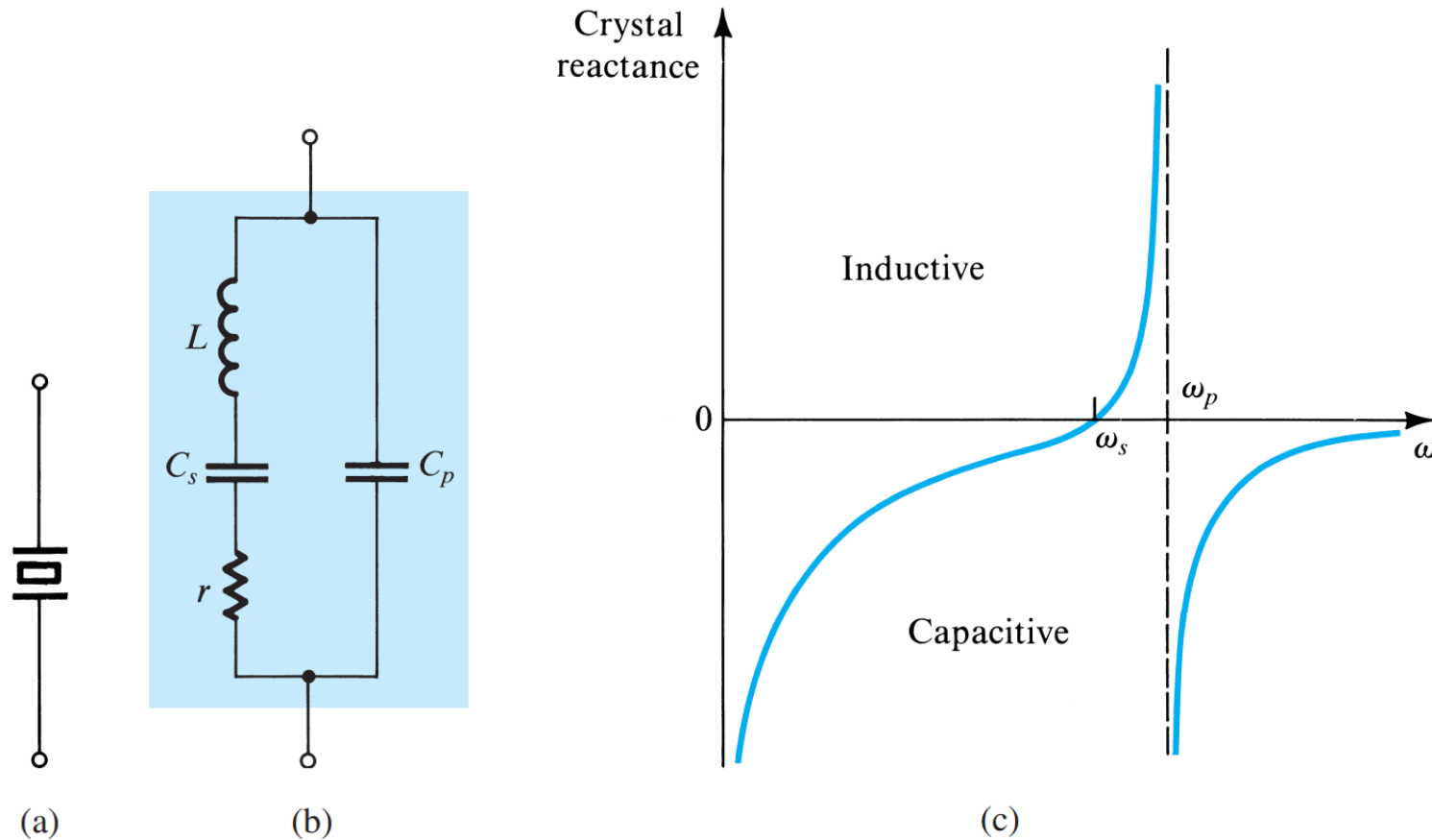


Figure 18.17 A piezoelectric crystal. (a) Circuit symbol. (b) Equivalent circuit. (c) Crystal reactance versus frequency [note that, neglecting the small resistance r , $Z_{\text{crystal}} = jX(\omega)$].

$$Z(s) = 1 / \left[sC_p + \frac{1}{sL + 1/sC_s} \right]$$

$$Z(s) = \frac{1}{sC_p} \frac{s^2 + (1/LC_s)}{s^2 + [(C_p + C_s)/LC_s C_p]}$$

a series resonance at ω_s

$$\omega_s = 1/\sqrt{LC_s}$$

and a parallel resonance at ω_p

$$\omega_p = 1 / \sqrt{L \left(\frac{C_s C_p}{C_s + C_p} \right)}$$



BANGLADESH
UNIVERSITY OF
ENGINEERING &
TECHNOLOGY

DEPARTMENT OF **E**LECTRICAL &
ELECTRONIC
ENGINEERING

Thank You!

If you have any questions, feel free to ask.